Auto Installer Genius

Automated Remote Software Installation Tool

User Manual | Version 1.0

Version	1.0
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Platform	Windows (local) to Ubuntu 22.04 (remote via SSH)
AI Engine	Cerebras | qwen-3-235b-a22b-instruct-2507
Repository	https://github.com/vishnu-18/auto-installer

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1. Overview

Auto Installer Genius is a Windows desktop application that automates the installation and removal of software on remote Ubuntu machines via SSH. It uses the Cerebras AI API to intelligently generate shell commands, verify their success, and automatically fix failures without any manual intervention.

Key Features

- AI-powered command generation using Cerebras (qwen-3-235b model).
- Automatic SSH connection with non-interactive sudo support.
- Pre-flight check - detects if software is already or partially installed.
- Automatic cleanup of partial installations before a fresh install.
- Real-time execution log with colour-coded output.
- AI verification of every command - auto-fix and retry on failure.
- Install, Uninstall, Pause, Resume, Cancel, and Terminate controls.
- Save and load connection configurations (.mc.json files).
- Local package upload - install from local files instead of downloading.
- Configurable log file directory.

2. Application Layout

[Screenshot: Full application main window]

Fig 2.1 - Auto Installer Genius main window

The application window has four main sections:

Section	Description
Input Panel	Enter software name, remote host IP, username, and password.
Control Buttons	Install, Uninstall, Pause, Resume, Cancel, Quit, Terminate.
Progress Bar	Shows real-time progress: X / N commands completed.

Execution Log	Scrollable log of all commands, outputs, and AI verdicts.

3. Input Fields

Fill in all four fields before clicking Install or Uninstall:

Field	Description	Example
Software Name	Name of the software to install/uninstall	New Relic
Remote IP / Host	IP address or hostname of target machine	192.168.100.197
Username	SSH login user (must have sudo access)	ubuntu
Password	SSH login password	**********

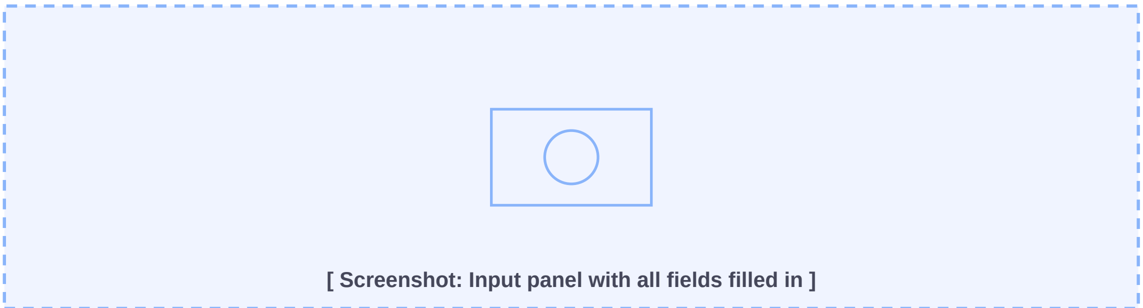

Fig 3.1 - Input panel with connection details entered

Note: The remote OS is detected automatically after SSH connects. You do not need to specify it manually.

4. Installing Software

To install software on the remote machine:

- **Step 1:** Fill in all four fields in the Input Panel.
- **Step 2:** Click the green **Install** button.
- **Step 3:** Watch the Execution Log for real-time progress.
- **Step 4:** A success dialog appears when installation completes.

The application performs these steps automatically in the background:

Step	Action
1	Fill in all four fields in the Input Panel.
2	Click the green Install button.
3	The app connects via SSH and detects the remote OS.
4	Stale apt sources (elastic, newrelic) are cleaned up automatically.
5	Pre-flight check runs - AI determines if software is already installed.
6	If partially installed, a cleanup runs before the fresh install.

7	AI generates the full list of installation commands.
8	Each command is executed on the remote machine one by one.
9	After each command, AI verifies success or failure.
10	On failure - AI generates fix commands and retries.
11	On fix failure - AI tries a completely different approach.
12	A success dialog appears when all commands complete.

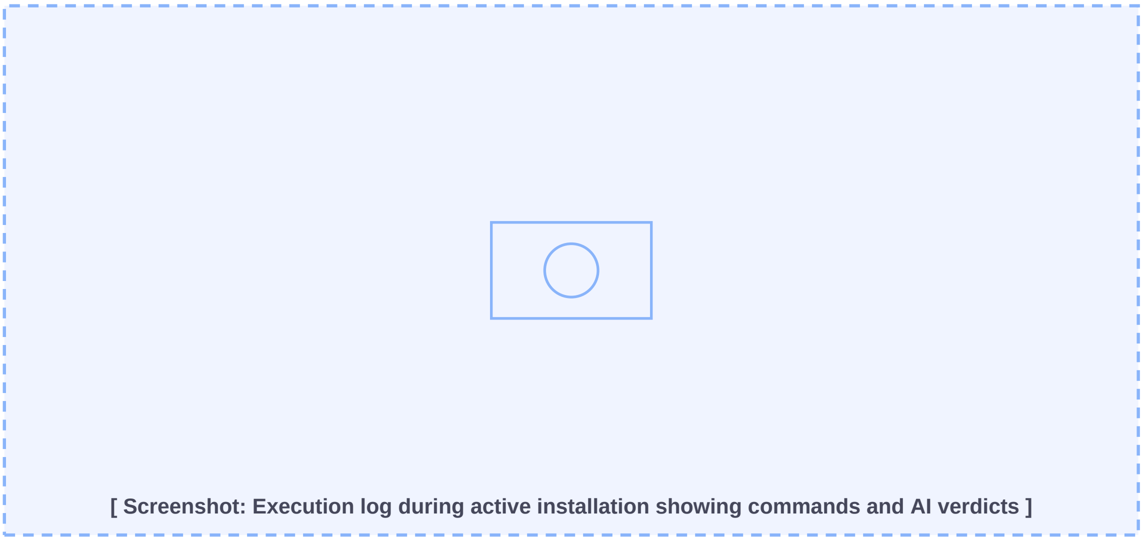

Fig 4.1 - Execution log during installation

5. Uninstalling Software

To completely remove software from the remote machine:

- **Step 1:** Fill in the Software Name, Host, Username, and Password.
- **Step 2:** Click the orange **Uninstall** button.
- **Step 3:** Confirm the dialog - 'This will completely uninstall...'
- **Step 4:** The app connects, fixes apt sources, then asks AI for uninstall commands.
- **Step 5:** Each uninstall command is executed and AI-verified.
- **Step 6:** A success dialog confirms the software has been removed.

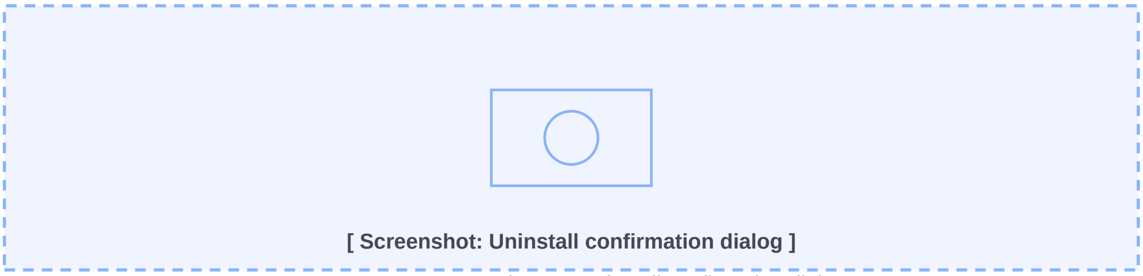

Fig 5.1 - Uninstall confirmation dialog

Note: The Uninstall flow asks the AI for commands based on the software name alone. It does not require a prior installation history in the current session.

6. Monitoring Progress

The Progress section shows a progress bar and a command counter (e.g. 3 / 8 commands). The Execution Log uses colour coding to make it easy to follow what is happening:

Log Colour	Meaning
Yellow	Command being executed (lines starting with >>>)
Green	Success message or AI SUCCESS verdict
Red	Failure message or AI FAILURE verdict
Blue	Info messages: Connecting, AI asking, Detected OS, etc.
White	Raw command output from the remote machine

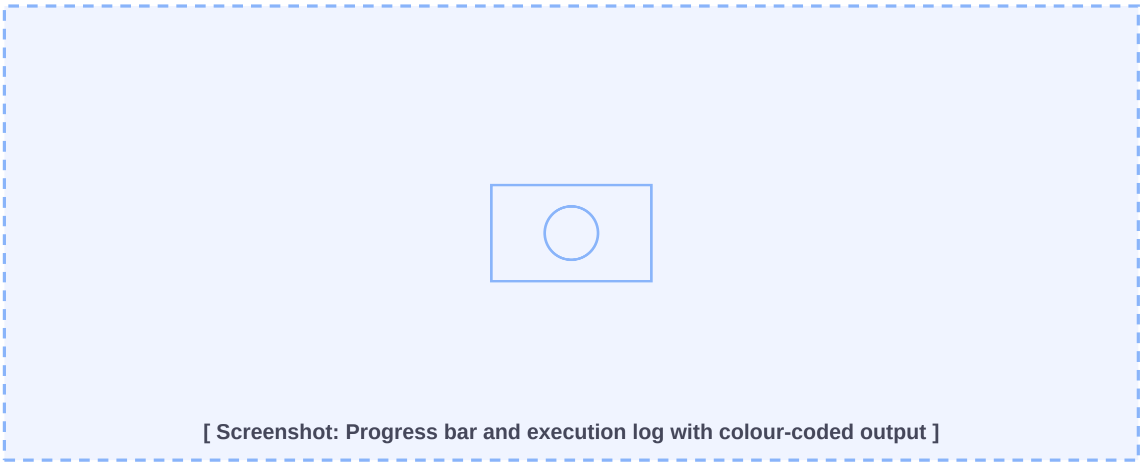

Fig 6.1 - Progress bar and colour-coded execution log

7. Pausing and Resuming

Click the yellow **Pause** button at any time during installation to pause after the current command finishes. The button label changes to **Resume**.

Click **Resume** to continue from where it left off.

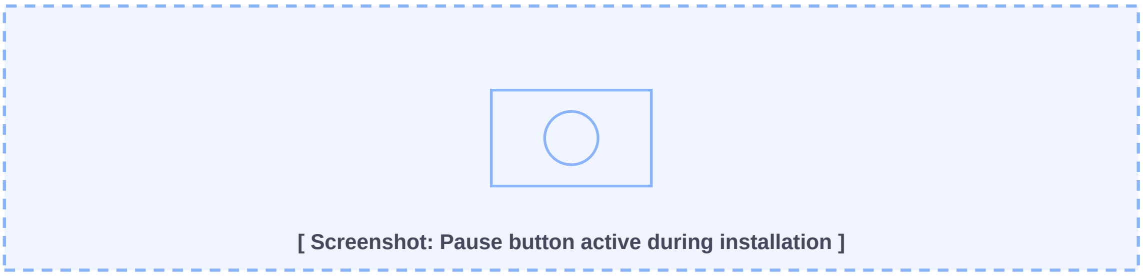

Fig 7.1 - Pause button during an active installation

Note: Pause takes effect after the currently running command and its AI verification finish. It does not interrupt a command mid-execution.

8. Cancelling an Installation

Click the red **Cancel** button to abort the installation. The application will:

- Wait for the current command to finish.
- Ask the AI for uninstall/rollback commands based on what was already executed.
- Run the rollback commands to clean up the partial installation.
- Show status as 'Cancelling - Uninstalling...' during rollback.

[Screenshot: Cancel confirmation dialog]

Fig 8.1 - Cancel confirmation dialog

*Note: Cancel is a graceful rollback. Use the red **Terminate** button only for an emergency hard stop - it exits immediately without any cleanup or rollback.*

9. File Menu - Saving and Loading Configurations

Connection details can be saved to **.mc.json** configuration files so you do not have to re-enter them each time.

[Screenshot: File menu open showing New, Open, Save, Recent options]

Fig 9.1 - File menu

Menu Item	Shortcut	Description
New	Ctrl+N	Clear all fields and start fresh.
Open...	Ctrl+O	Load a saved .mc.json configuration file.
Save	Ctrl+S	Save current fields to the current file.
Save As...	Ctrl+Shift+S	Save to a new .mc.json file.
Recent Configurations	-	Quick-open recently used config files.
Quit	-	Exit the app (waits for current command).

Note: Passwords are base64-encoded in saved config files. Do not share .mc.json files that contain credentials.

10. Settings Menu

The Settings menu has three options:

10.1 Environment Variables (.env)

Opens a text editor showing the contents of the .env file. Edit your Cerebras API key and other variables here, then click **Save**. The file is reloaded immediately - no restart required.

[Screenshot: Environment Variables editor dialog with Save and Cancel buttons]

Fig 10.1 - .env editor dialog

10.2 Log File Path

Choose the directory where installation log files are saved. Click **Browse...** to pick a folder, then **Save**. The setting persists across sessions in .settings.json.

[Screenshot: Log File Path dialog with Browse button]

Fig 10.2 - Log File Path configuration dialog

10.3 Local Package Directory

If you have package files already downloaded on your Windows machine (e.g. .deb files, .tar.gz archives), set this directory path here. When set, all files in that directory are uploaded to the remote machine via SFTP before installation begins. The AI is then instructed to use those local files instead of downloading.

[Screenshot: Local Package Directory dialog with Browse and Clear buttons]

Fig 10.3 - Local Package Directory configuration dialog

Note: Leave Local Package Directory empty to use the default behaviour where the AI downloads packages from the internet on the remote machine. Click Clear to reset it.

11. Log Files

Every install and uninstall operation is automatically saved to a log file. Log files are named:

```
install_SoftwareName_YYYYMMDD_HHMMSS.log
```

Default location:

```
F:\MySws\auto-installerV1.0\logs\
```

Each log file contains:

- Full path and timestamp of the log file.
- Software name, host, and start time.
- All AI-generated commands (numbered list).
- Raw output from each command on the remote machine.
- AI verdict for each command (SUCCESS / FAILURE + reason).
- Fix commands and alternative commands if triggered.
- End timestamp.

Note: Change the log directory via Settings - Log File Path. The directory is created automatically if it does not exist.

12. Buttons Reference

Button	Colour	State	Description
Install	Green	Always	Start installation of the specified software.
Uninstall	Orange	Always	Completely remove the specified software.
Pause	Yellow	Running	Pause after the current command finishes.
Resume	Yellow	Paused	Continue a paused installation.
Cancel	Red	Running	Abort and rollback via AI-assisted uninstall.
Quit	Gray	Always	Exit app gracefully (waits for current command).
Terminate	Bright Red	Always	Emergency hard exit - no cleanup, immediate.

[Screenshot: Control buttons bar showing all buttons]

Fig 12.1 - Control buttons bar

13. Understanding AI Verdicts

After each command executes, the AI analyses the output and exit code to determine if it succeeded. The verdict appears in the log as:

```
AI verdict: SUCCESS - The package was installed successfully.
AI verdict: FAILURE - apt-get returned a non-zero exit code.
```

Verdict	Meaning	What Happens Next
SUCCESS	Command completed as expected.	Move to the next command.
FAILURE	Command failed or produced errors.	AI generates fix commands.
Fix SUCCESS	Fix commands resolved the issue.	Re-run or skip original command.
Fix FAILURE	Fix commands also failed.	AI tries alternative approach.
Alt FAILURE	All approaches failed.	Installation aborted (FAILED).

The AI also short-circuits verification for idempotent results. If the output contains phrases like 'already installed', 'already the newest version', or 'nothing to do', it is automatically marked as SUCCESS without an API call.

14. Tips and Best Practices

- Always test with a non-production VM before running on live servers.
- Save your connection config (File - Save) to avoid re-entering details each time.
- Check the log file after each run - it contains the full command history.
- If installation fails, the log shows exactly which command failed and why.
- Use the Local Package Directory setting for environments where the remote machine has no internet access.
- The Cerebras free tier supports two models. The app uses the larger qwen-3-235b model for better command accuracy.
- Rotate your Cerebras API key periodically and update it via Settings - Environment Variables.
- Never share your .env file or .mc.json config files containing passwords.
- Use Pause if you need to inspect the remote machine mid-installation.
- Use Terminate only as a last resort - it does not clean up partial installs.